

LUNGSYNTHAI: PHASE-CONDITIONED DIFFUSION MODELS FOR 4D LUNG CT SYNTHESIS

AI-Based Reconstruction of Intermediate Lung CT Phases

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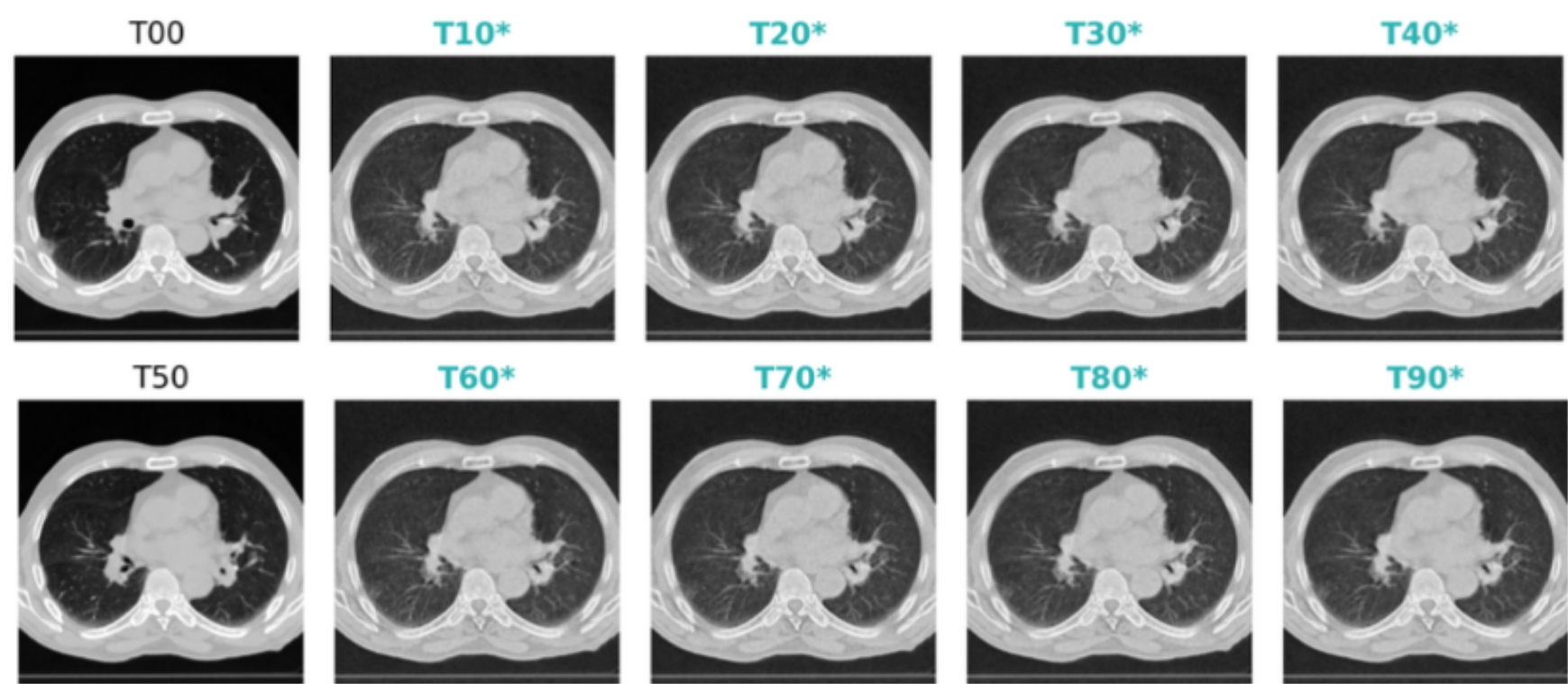
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1. Introduction

- 4D-CT captures lung motion across the breathing cycle, guiding radiotherapy planning.
- 10 phases per scan (T00–T90); only the boundary phases, T00 (exhale) and T50 (inhale), come out clean.
- The 8 intermediate phases are often blurry, which weakens tumour tracking and localisation.
- We propose a phase-conditioned 3D diffusion model that reconstructs those intermediate phases directly from the two clean boundary scans.

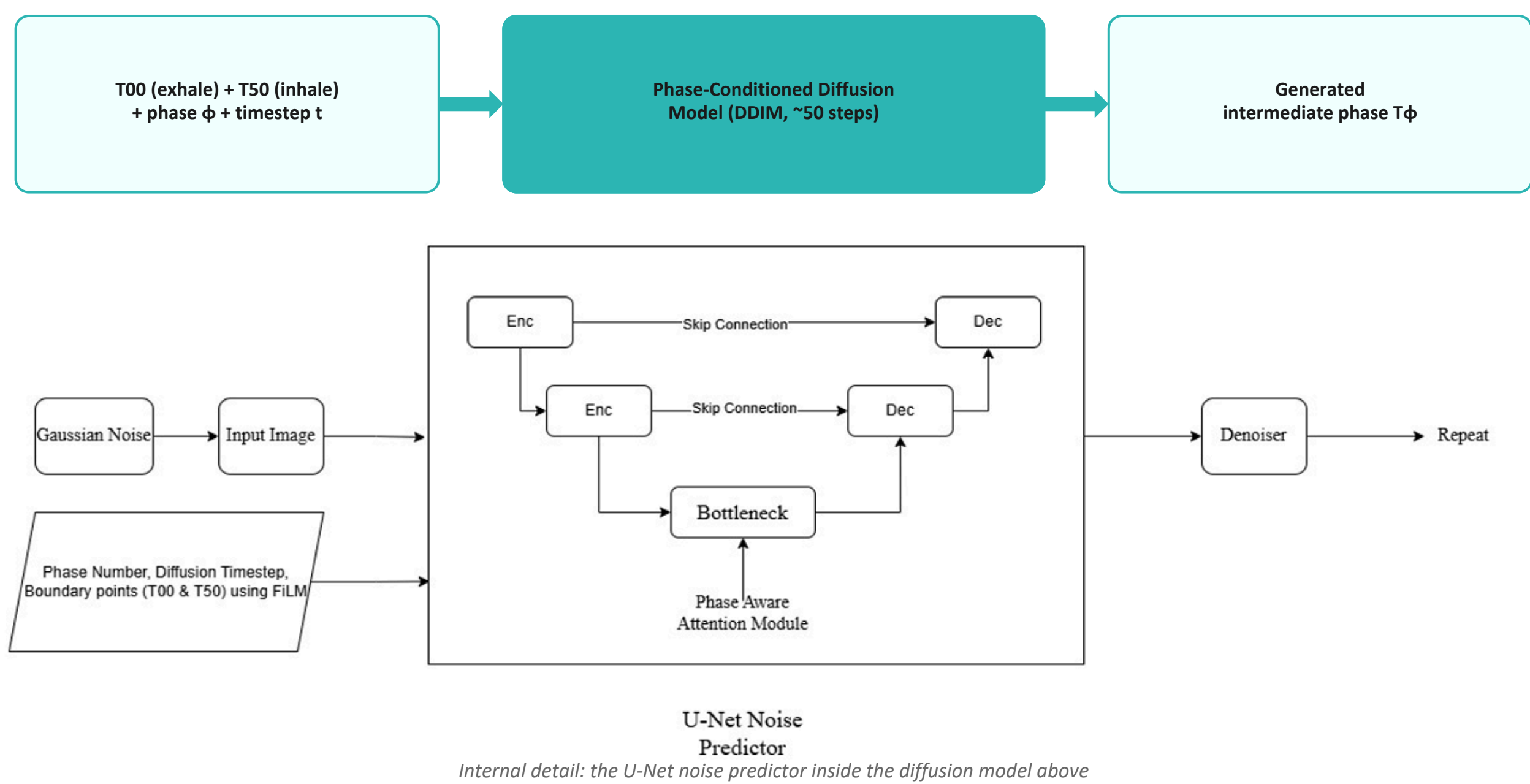


Full reconstructed breathing cycle for Case 3 — teal labels (T10–T90 except T50) are generated by the model; T00/T50 are the real anchor scans.

2. Background

- Conventional methods often rebuild intermediate phases from the degraded scans themselves, inheriting their blur and artifacts.
- Diffusion models offer a strong alternative — learning to reverse noise step by step for high-quality synthesis.
- We instead condition only on the two clean boundary scans (T00, T50), never touching a degraded intermediate scan.
- One model generates all 8 intermediate phases, guided by an explicit phase number.

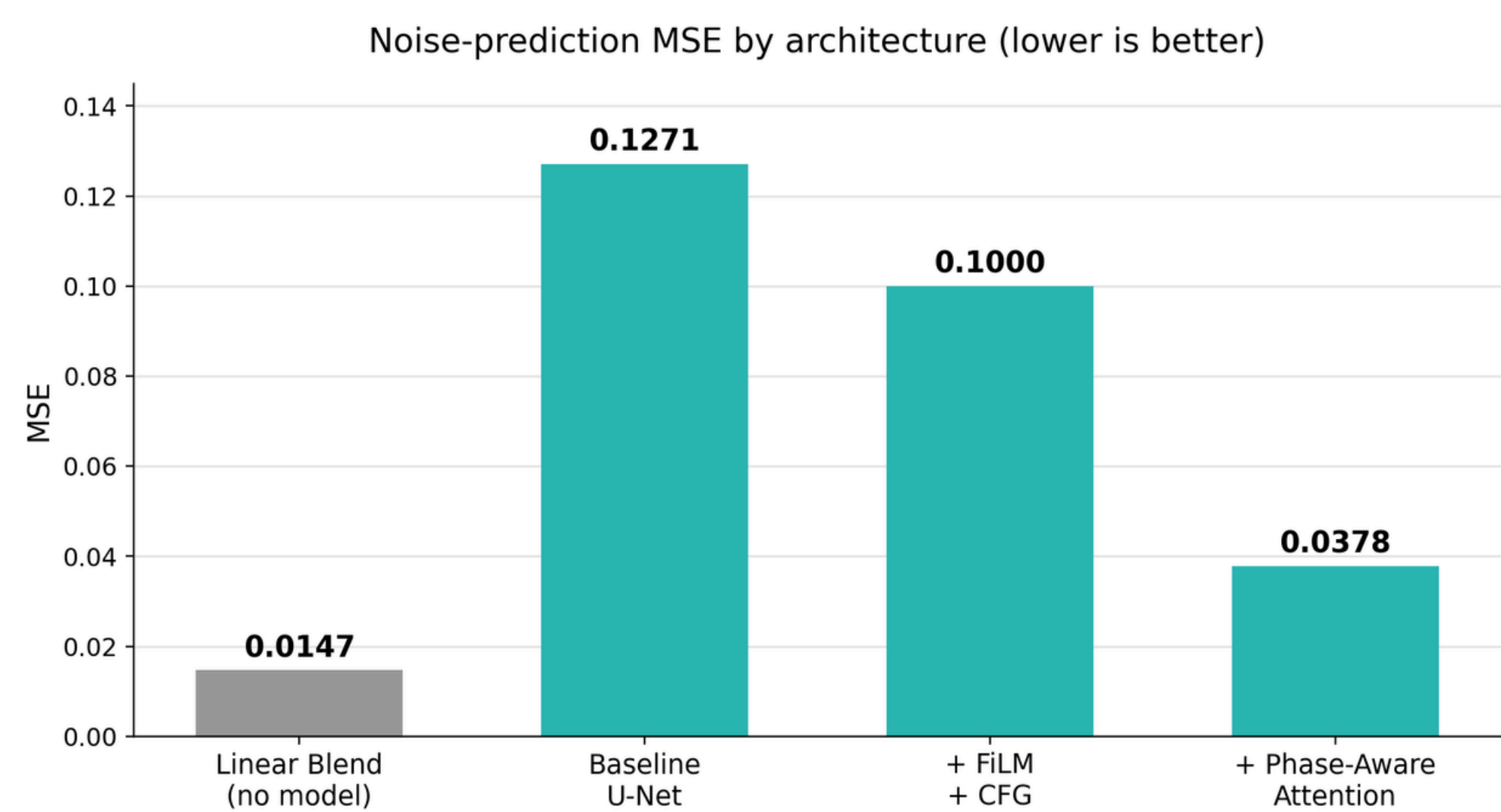
3. System Architecture



- 3D U-Net predicts the noise in a noisy intermediate phase, conditioned on T00 + T50 as extra input channels.
- Phase number and timestep are injected at every layer via FiLM.
- Phase-Aware Attention (at the bottleneck) focuses on the highest-motion regions for that phase.
- DDIM sampling generates each phase in ~50 steps — 20x faster than standard diffusion.

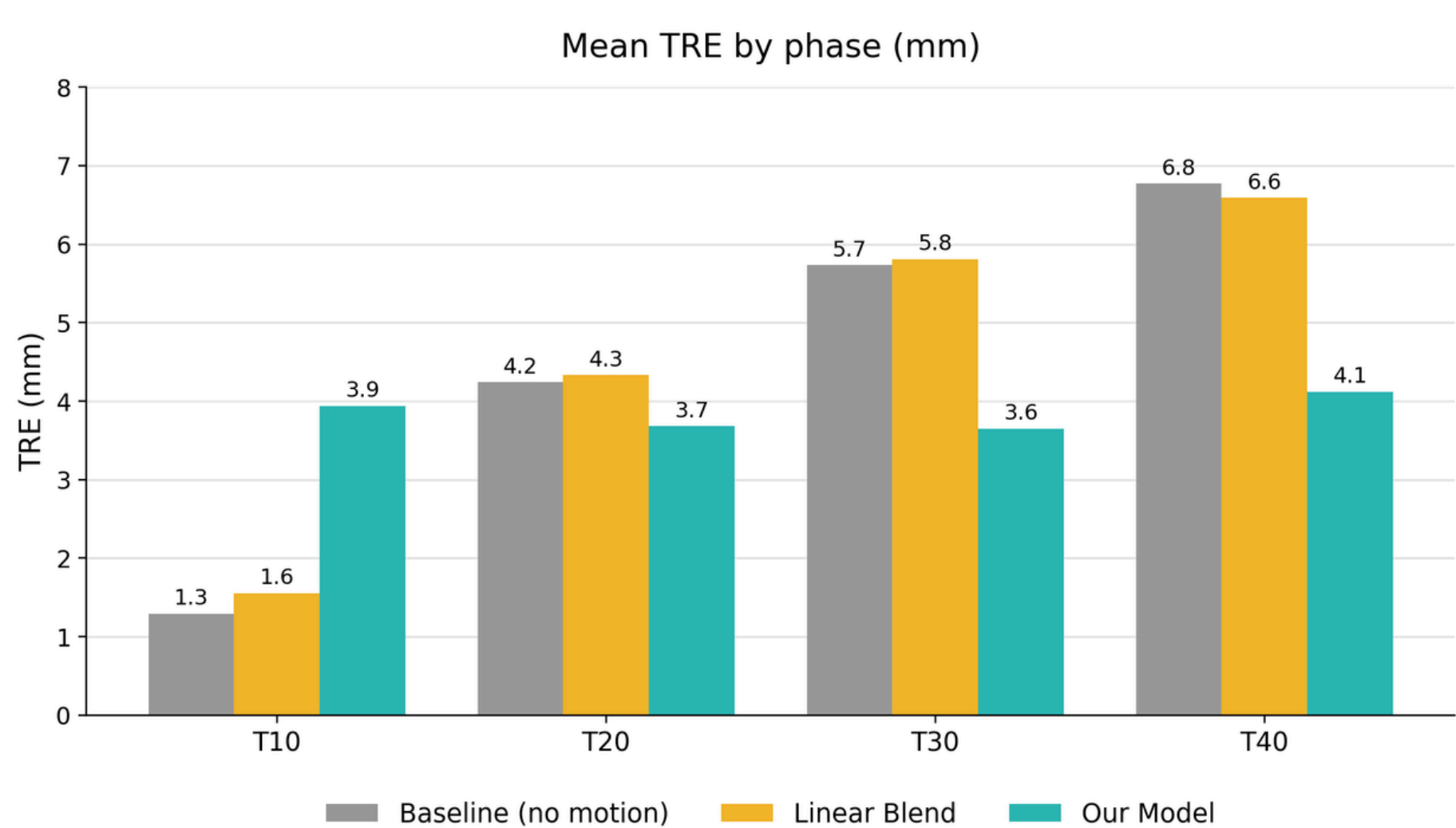
4. MSE (Training Ablation)

- MSE = error between predicted and actual noise during training — an internal signal, not a clinical metric.
- FiLM + classifier-free guidance, then Phase-Aware Attention, steadily reduce it.
- Caveat: a naive T00/T50 blend (no model) scores lower MSE than every trained model — MSE rewards blur, which is why TRE is our real metric



5. TRE (Clinical Evaluation)

- Evaluated on 2 held-out patients (Case 3, Case 9), phases T10–T40, 75 landmarks each.
- Model beats both baselines from T20 onward, slightly overshoots at T10. Mean TRE: 3.85mm (model) vs 4.50mm (no motion) vs 4.57mm (blend).
- Blending (4.57mm) barely beats doing nothing (4.50mm) — it shifts intensities without actually moving anatomy.



6. Conclusions

- Phase-conditioned 3D diffusion model generates intermediate lung CT phases from two boundary scans alone
- Evaluated on 2 held-out patients, T10–T40, 75 landmarks/phase. Mean TRE 3.85mm vs. 4.50mm no-motion baseline; gains concentrated at T20–T40 (largest true motion).
- Limits: 8–10 patient anatomies.
- Future work: larger multi-institutional datasets and web-interface deployment.

